

Referee Report and Computational Stress Test: the Round-12 Patchwork-Bag Automata Proof of Decidability for $\mathcal{ALCI}_{\text{RCC5}}$

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Abstract

This report reviews the canonical round-12 decidability manuscript `papers/automata_route_repairs/split_forest_au` (29 pp, May 31, 2026) and attacks it with three new computational probes (WP14–WP16, under `verification/python/`). Summary verdict: **genuine gap; the theorem remains most likely true; the repair is localized and computationally corroborated**. The round-12 architecture is a real improvement — the route-dependent transformer layer is gone, and global RCC5 coherence of the *bag networks* genuinely follows from the RCC5 patchwork property plus a correct compactness argument (WP14 verifies the patchwork statement exhaustively at small widths: 63,219 amalgamation instances, zero failures). But the Truth Lemma of Theorem B(b) contains one unproved load-bearing step: it assumes the declared relation-vector *shadows* agree with the patchwork completion ρ_T , although ρ_T is amalgamated from bag networks *only* and no listed legality test constrains two shadows against each other. WP15 exhibits a three-bag configuration that passes every listed test yet admits *no* frame realizing all declared vectors (23 of 64 assignments on the minimal pattern fail); the missing statement is a *Shadow Amalgamation Lemma*, the round-12 successor of the sixth review’s transformer-coherence gap C1 — same genus, one layer up, but now much smaller. Two independent specification loopholes (EQ-valued shadow entries; undefined shadow scope) are each shown by WP16 to flip the referee concept C_{split} and three other unsatisfiable forced-composition concepts to ACCEPT; both have one-line fixes. Under the repaired discipline, WP16’s mechanism agrees with the cover-tree tableau on all ten diagnostics — including the strong-EQ PO-loop concept that the retired quasimodel reasoner decided wrongly — and on a 148-concept random sweep with zero disagreements. No accepted-but-unsatisfiable *concept* was found. Status recommendation: keep “strongly supported, *not* certified”; the keystone is now the Shadow Amalgamation Lemma plus the $K(C_0)$ finiteness accounting.

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*Seventh review in the series. Prompted by Michael Wessel. This is a *context-loaded* (“hot”) adversarial audit, *not* a cold review in the sense of the project’s cold-review protocol: the reviewer had full access to the repository, including all prior referee reports. It is, however, by a different model lineage than the author of the manuscript under review (GPT-5.5, OpenAI).

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1 Scope, materials, and disclosure

Manuscript under review. `split_forest_automata_with_appendices.tex` (“the manuscript”), the round-12 *expanded self-contained edition*: Theorem A (split-forest normal form) $\rightarrow$ Theorem B (finite patchwork-bag abstraction adequacy) $\rightarrow$ Theorem C (two-way alternating parity tree automaton correctness) $\rightarrow$ decidability, with appendices A–U. The 14-page theorem-spine edition `split_forest_patchwork_automata_repaired.tex` was read for cross-checking; the sixth cold review (`referee_report_ALCI_RCC5.md`) and the author’s response (`formal_response_referee_round12.tex`) were used to fix the review targets.

Review goal. Attack the proof; search for counterexamples; where no defect is found, strengthen the verification record. Accordingly this report contains both findings (Section 3) and a list of steps checked and found sound (Section 4), plus three new verification work packages (Section 5).

Disclosure. This is a *hot* review: the reviewer had the full repository in context, including all six prior reviews. It therefore does not discharge the project’s requirement that round-12 be cold-reviewed; it is best read as an internal audit. The reviewer is a different model lineage (Anthropic) from the manuscript’s author (GPT-5.5, OpenAI).

2 Summary verdict

1. **The architectural move of round-12 is correct and real progress.** Deleting the pair/triple transformer layer and resting global RCC5 coherence on complete atomic bag networks + the patchwork property removes the sixth review’s C1 *for the bag layer*: Appendix A’s finite tree-patching induction is sound (its use of running intersection is exactly right), the compactness passage to countable trees is correct (modulo a wording slip, Finding 5), and the two-bag patchwork property itself — the one algebraic input — holds in exactly the stated form for the abstract semantics, now machine-verified exhaustively at small widths (WP14, 63,219 instances, zero failures).
2. **The Truth Lemma of Theorem B(b) has one unproved load-bearing step (Finding 1, critical).** ρ_T is amalgamated from *bag networks only* (App. E, “Global realization”), yet the universal case asserts “the exact relation vector from x to the live ports includes $\rho_T(x, y) = R$ ”. Nothing forces the declared shadow vectors to agree with ρ_T : no listed test relates two shadows to each other, and a vector entry to a port is dropped when the port is forgotten. WP15 gives a three-bag configuration passing *every* listed test whose declared vectors are *jointly unrealizable* — the quoted sentence is false for every choice of ρ_T on that input. The missing statement is a *Shadow Amalgamation Lemma*. This is the same *genus* of defect as the sixth review’s C1 (local checks that do not compose globally), one layer up (shadow tier instead of transformer tier) — but the layer is far thinner, the failure is now precisely isolated, and two repair disciplines are computationally corroborated (Section 6).

3. **Two specification loopholes are independently soundness-fatal as literally written** (Findings 2 and 3); each admits a one-line fix, and WP16 machine-checks that with the fixes in place the mechanism decides all referee diagnostics correctly, while either loophole alone flips C_{split} , C_{force} and two deeper forced-composition concepts to ACCEPT.
4. **The $K(C_0)$ finiteness argument is still an assertion** (Finding 4) — the sixth review’s M1 in softened form.
5. **No accepted-but-unsatisfiable concept was found.** Like the sixth reviewer, I could not exploit the gaps at *concept* level: $\mathcal{ALCI}$ lacks nominals, so the witness-identification needed to force two distinct occurrences into my WP15 configuration appears inexpressible. The configuration-level defect stands regardless: the Truth Lemma, as proved, is false.
6. **Status recommendation.** Keep “strongly supported, *not* certified.” Do not describe round-12 as cold-reviewed. The keystone has narrowed again: from “patchwork-bag adequacy” generically to *Shadow Amalgamation* + $K(C_0)$ *accounting* specifically.

3 Findings

Throughout, Comp is the RCC5 composition table (re-verified, S1), and “the listed tests” means the union of Definition 3.4 (V1)–(V8), the Appendix H shadow-movement conditions (boundary agreement on shared ports; compatibility with the current bag network), and the Appendix Q tests (N1)–(N6), (T1)–(T4), (W1)–(W3), (O1)–(O5), (C1)–(C3), (S1)–(S4).

Finding 1 (CRITICAL — the Truth Lemma assumes an unproved Shadow Amalgamation Lemma). *Location:* Theorem 3.6(b) proof; Appendix E (“Global realization”, “Truth lemma”); Appendix M (“Truth lemma: universal cases”, “Truth lemma: existential cases”).

The step. The realization builds ρ_T by “the countable patching lemma ... applied to the tree-shaped family of finite *bag networks*” — shadow vectors are not part of the amalgamated family. The universal case then argues: when target y is introduced, “the exact relation vector from x to the live ports includes $\rho_T(x, y) = R$; validity therefore forces D into the type of y .” The existential case for vertical requests likewise fulfills $\exists R.D$ at “a port q whose exact recorded relation to the source is R ” — where, if the source is no longer live, “recorded” can only mean the shadow entry.

Why it does not follow. For a pair (x, y) that is never co-bagged, $\rho_T(x, y)$ is whatever the bag-only amalgam happens to choose among the (generally several) patchwork-feasible values, while the declared vector entry is a separate guess constrained only *per bag* (one-point compatibility) and *per boundary* (rigidity on continued ports). Two gaps result:

- (a) nothing makes ρ_T agree with the declared entries; and
- (b) one cannot repair this by amalgamating the entries too, because the listed tests never constrain two *shadows* against each other, nor a vector entry declared while a port was live against entries declared after that port is forgotten. The implicit lemma — *bags + all declared shadow rows are jointly realizable in one frame* — is false for the tests as written.

Machine-checked counterexample (WP15, part P1; three bags on a decomposition path, one port each, separators empty — a variant with a continued all-DR separator port behaves identically):

bag	live port	declared shadow entries at this bag
t_0	x_1	— (x_1 introduced, then forgotten)
t_1	x_2	x_1 : ($x_2 \mapsto \text{DR}$)
t_2	p	x_1 : ($p \mapsto \text{PP}$); x_2 : ($p \mapsto \text{PPI}$)

Every bag is (trivially) a complete atomic closed network; every row is one-point composition-closed with its bag; rigidity is vacuous (no continued ports); all entries are non-EQ; the (x_1, x_2) entry even respects the residual-frontier discipline (DR). Yet jointly the three declarations say $x_1 \text{ PP } p$, $p \text{ PP } x_2$ (converse of the declared PPI), hence $\rho(x_1, x_2) \in \text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$, contradicting the declared DR. **No abstract RCC5 frame realizes bags + rows**; consequently, for every choice of ρ_T , at least one declared vector entry is wrong, and the Truth Lemma’s quoted sentence fails: an obligation $(\rho_T(x_1, x_2), D) \in \Omega_{x_1}$ can fire in the realized frame on a target whose type was never checked for D (the automaton checked the DR entry instead). WP15’s exhaustive sweep of the pattern shows the defect is generic: **23 of the 64** legal row assignments are jointly unrealizable.

What survives. With a *single* shadow the implicit lemma is provable and empirically clean (WP15 P3: 250/250 realizable): the extended family $\{N_t \cup \text{row}_t\}$ is itself a tree-shaped agreeing family of complete atomic networks (the source becomes a member of every bag in its connected scope; rigidity is exactly overlap agreement), so Theorem 2.5 applies to it verbatim. The defect is strictly about *multiple* shadows and *dropped* entries. This localizes the missing lemma sharply; see Section 6.

Relation to prior reviews. This is the successor of the sixth review’s C1: round-12 removed the route-dependent transformer layer, but the shadow layer still carries cross-bag relational commitments whose global coherence is asserted rather than derived from patchwork. The manuscript’s own Appendix U names the failure mode (F3: “if relation-vector shadows failed to preserve universal obligations across bag boundaries, the automaton could accept false abstractions”).

Finding 2 (MAJOR (spec, soundness-fatal as written, one-line fix) — EQ-valued shadow entries are not excluded). *Location:* Definition 3.3 ($\vec{r} \in \text{Rel}^P$ with $\text{Rel} = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}$); Appendix Q (no test excludes EQ entries; the no-EQ tests (N3)/(N4) apply to bag networks only).

A shadow whose entry to a target port is declared EQ triggers *no* obligation (R, D) with $R \neq \text{EQ}$, while under strong equality the realized relation between the two distinct occurrences is necessarily non-EQ. On the referee concept $C_{\text{split}} = \exists \text{PP}. (A \sqcap \neg B \sqcap \forall \text{PP}. \neg B \sqcap \forall \text{PPI}. \neg B \sqcap \forall \text{PO}. \neg B) \sqcap \exists \text{PP}. B$ the y -shadow may declare its entry to z as EQ; all three universal obligations then stay silent, and the abstraction is accepted although $\text{Comp}(\text{PPI}, \text{PP}) = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}\}$ minus EQ leaves only obligation-violating options. WP16 (literal mode) machine-checks the acceptance. *Fix:* require shadow entries to range over $\text{Rel} \setminus \{\text{EQ}\}$ (an entry EQ is legal only as the statement that the source *is* that port’s occurrence, i.e. never for a genuine shadow).

Finding 3 (MAJOR (spec, soundness-fatal as written) — the shadow’s scope is never pinned down). *Location:* Appendix H (“universally branches to all adjacent bags in which the shadow *must* continue” — “must” is not defined); Definition 3.4 (V5) (“every introduced target is checked” — by which shadow, where?); Appendix S.

If a run may simply not carry the shadow of x into the subtree where y lives, the pair (x, y) escapes all checks; the completeness direction never objects (a real model needs no check), so the choice is unconstrained. WP16 (literal mode) shows the consequence: the unsatisfiable C_{force} , C_{split} , $C_{2\text{hop}} = \exists \text{DR}. (\exists \text{PPI}. A) \sqcap \forall \text{DR}. \neg A$ and $C_{3\text{hop}} = \exists \text{DR}. (\exists \text{PPI}. \exists \text{PPI}. A) \sqcap \forall \text{DR}. \neg A$ are all ACCEPTED when distant pairs escape obligations. Note that mandatory propagation cannot be depth-bounded: $\text{Comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$ forces a DR-obligation into *every* proper part of a DR-witness, arbitrarily

deep, so obligation-carrying shadows must be propagated into entire subtrees. That is unproblematic for an alternating automaton (spawn a copy per direction forever; the state space is the finite set of profiles) — but it must be *required* by the transition function, not left to the run. *Fix*: make shadow propagation mandatory: for every source with a nonempty obligation set, a shadow copy is spawned into every child direction and to the parent, at every bag of the run.

Finding 4 (MAJOR — $K(C_0)$ finiteness is still asserted (sixth review M1, softened)).

Location: Lemma 3.5 (1em:K-bound) and Appendix K (K1)–(K4).

The rank-0 step says: “one enumerates finite labelled contexts, keeps those satisfying local RCC5 consistency and saturation, and quotients by isomorphism. . . This is finite.” As written it is not: the saturation rules (S1)–(S6) are conditional-insert closure rules, so a context with arbitrarily many mutually-DR ports (all conditions vacuously satisfied) passes them at every size, and isomorphism classes grow without bound. The enumeration in Appendix K terminates only if saturated contexts *needed for* C_0 have bounded minimal representatives — which is precisely the claim to be proved. The round-11 recurrence was faulted for over-claiming a specific bound (M1); round-12 responds by claiming no bound at all, which under-delivers: *some* explicit accounting is needed. *Fix*: define side contexts as *witness-generated* (every port carries a generation reason: a selected witness of a closure formula at a specific source, a boundary insertion under (S2), a stratum/threshold node under (S3)/(S4), or a recursive-context port under (S5)); then bound reasons per source by $|\text{Cl}(C_0)|$ plus the boundary size and prove the closure operator stays within a computable per-rank recurrence — or sidestep entirely with the $2^{\text{poly}(|\text{Cl}(C_0)|)}$ type/mosaic bound the sixth review recommended.

Finding 5 (MINOR — Appendix A countable-patching wording). The closed sets are defined as labellings “extending *the* patchwork completion on the occurrences occurring in S ” — there is no canonical completion, and with the definite article the finite-intersection argument breaks (different subtrees would pin incompatible completions). Replace by “whose restriction to the occurrences of S is *some* complete atomic composition-closed extension of all bags in S ”. With that reading the argument is correct (checked in detail): the sets are clopen cylinders, a common finite super-subtree witnesses the finite intersection property (restriction of a completion of the larger subtree to a smaller one remains a completion because (R1)–(R3) are universal conditions), and every pair/triple of occurrences lies in some finite connected subtree by trace-connectedness, so the limit labelling satisfies all constraints.

Finding 6 (MINOR — parity narrative and input-tree formalities). (i) The “generalized Büchi via round-robin counter” construction is the single-thread device; in an *alternating* automaton each spawned request copy can simply carry a rejecting (odd) priority until it enters its fulfilled state, which is simpler and avoids any inter-request bookkeeping. Either is fine; the text should commit to one and state the priority assignment. (ii) “Runs on ranked trees over Σ_{C_0} ”: bag labels do not determine an arity; say explicitly that trees are padded to a fixed maximal branching degree (computable from K and $|\text{Cl}(C_0)|$) with a dummy label. (iii) Definition 3.4 (V8) ties a vertical request to “the same overlap-equivalence class of the source occurrence”; if this is meant to keep the source *live* (co-bagged) along its support branch — which makes request fulfillment a bag-network fact and immunizes it against Finding 1 — say so explicitly; it is the right choice.

Finding 7 (OBSERVATION — the parity/request-closure content is load-bearing (incidental replication of the round-6/7 defect)). While building WP16’s reference machine, a first implementation used naive same-type blocking for vertical eventualities. It promptly produced a false SAT on $C_{3\text{hop}}$ by deferring the eventuality $\exists \text{PPI.A}$ forever down the regular tail — precisely the lap-collapse family of defects from the no-automata thread (rounds 6/7), and precisely what a

parity condition must reject. The fix (a lap may block only if the blocking occurrence fulfilled the *same* request concretely) restored agreement with the tableau. This is independent corroboration that Theorem C’s request states must enforce fulfillment *on every lap*, and that any implementation gets this wrong by default.

4 Steps attacked and found sound

S 1 (Composition table). Re-derived independently from set semantics over a 5-element universe (WP14 T0); matches the manuscript’s table in all 25 cells.

S 2 (The patchwork theorem as stated (Theorem 2.5)). For the abstract composition-table semantics, “satisfiable finite complete atomic network” = “composition-closed network”, so the two-bag patchwork property is a purely combinatorial statement. WP14 verifies it **exhaustively** for one fresh variable per side with shared parts of size 0..3 (21,538 agreeing pairs) and for two fresh variables per side with shared parts of size 1..2 (41,681 pairs, sampled at the largest class), and verifies the finite tree-patching lemma on 300 random tree-shaped bag families (3–6 bags, width ≤ 7) by literally executing Appendix A’s leaf-by-leaf induction. **Zero failures.** This turns the proof’s central external dependency into a machine-checked fact at small width (complementing WP6), in *exactly* the form the manuscript uses it.

S 3 (Finite tree patching’s use of running intersection). The induction step needs the leaf to agree with the *remainder* on their full shared variable set; connected traces give $V_L \cap V_{\text{rest}} = V_L \cap V_M$ for the unique neighbor M , which is exactly what the proof uses. Correct.

S 4 (Strong equality / Appendix G). Correct given atomic completions: every completion in the amalgam is atomic with EQ only on the diagonal, and no cell of the composition table can force a cross pair to be exactly EQ — WP14 T4 verifies that every nontrivial cell containing EQ also contains PP, PPI, PO. Together with exact occurrence identity this cleanly eliminates the historical blocking-vs-equality confusion (Appendix R’s trace-connectedness argument is also correct).

S 5 (The strong-EQ loop concepts). The mechanism (exact occurrence identity + same-bag witness reuse) decides the PO-loop concept $C \sqcap \exists \text{PO}.\exists \text{PO}.C \sqcap \forall \{\text{PO}, \text{DR}, \text{PP}, \text{PPI}\}.\neg C$ correctly as SAT (WP16), the known blind spot of the retired quasimodel reasoner. The universal constraints force the second-hop witness to be the root occurrence itself, and reuse expresses exactly that. This is an architectural strength of round-12.

S 6 (C_{split} under the repaired discipline). With Findings 2/3 fixed, the rejection mechanism works as Appendix O claims: the shared source x keeps y, z connected through its overlap class; the y -shadow’s entry to z is triangle-constrained through x to $\text{Comp}(\text{PPI}, \text{PP}) \setminus \{\text{EQ}\} = \{\text{PP}, \text{PPI}, \text{PO}\}$, and each option violates one of y ’s universals against $z:B$. WP16 confirms REJECT in repaired mode, ACCEPT under either loophole. The sixth review’s C2 is thus *addressed conditional on* Findings 1–3.

S 7 (Forest-to-tree packaging (sixth review C3.3)). Dissolved by design: the decomposition tree is a data structure, not the containment order; vertical direction is a per-edge annotation, upward growth is a child bag, and the cross-tree links of the semantic forest are side-context bags. No residual objection.

S 8 (Two-wayness and emptiness). Choosing a two-way alternating parity tree automaton and citing emptiness (Vardi) is the right resolution of the sixth review’s M3; the lossless one-way summary burden is genuinely gone. The state families are finite given the profile alphabet. (Presentational points in Finding 6.)

S 9 (Witness thinning; types; normalization). Lemma 3.1’s complement-closure argument is correct and standard (as in prior reviews); the EQ-normalization and Hintikka machinery are correct.

S 10 (Random cross-validation of the repaired mechanism). WP16’s repaired-discipline machine agrees with the cover-tree tableau on 148/148 decided random concepts (2 skipped at budget), in addition to all ten diagnostics. No adequacy anomaly surfaced.

5 Computational probes (new work packages)

All three scripts are self-contained except WP16’s use of the repo’s concept classes and tableau oracle; all pass; total runtime a few minutes.

WP14 — `wp14_patchwork_property_check.py`

Direct verification of Theorem 2.5 for the abstract semantics. T0: composition table re-derivation (PASS). T1: two-bag patchwork, exhaustive, 1+1 fresh, $|S| \leq 3$: 21,538 agreeing pairs, all amalgamable (PASS). T2: 2+2 fresh, $|S| \in \{1, 2\}$: 41,681 pairs, zero failures (PASS). T3: 300 random tree-shaped families amalgamated leaf-by-leaf per Appendix A (PASS). T4: EQ-forcing impossibility (PASS). *A failure here would have invalidated the proof (Appendix U, F1); none was found.*

WP15 — `wp15_shadow_row_realizability.py`

The Finding 1 attack and its repair map. P1: the minimal counterexample above passes all listed tests and is jointly unrealizable (also with a continued separator). P2: exhaustive mini-search over the pattern: **23/64** legal assignments unrealizable — the defect is generic, not a knife-edge. P3: **single-shadow** configurations: 250/250 realizable (the extended family $\{\text{bag} \cup \text{row}\}$ is an agreeing tree family, so Theorem 2.5 applies — the provable core of the repair). P4: multi-shadow with entries restricted to $\{\text{DR}, \text{PO}\}$ (the saturation/residual-frontier discipline applied to the shadow layer): **0** unrealizable in 350 random + 8 exhaustive configurations. P5: the extended-bag discipline (shadows as virtual ports with per-bag-checked shadow-shadow entries): **0** failures in 200 accepted configurations.

WP16 — `wp16_round12_bag_acceptance.py`

Acceptance-level mechanism check against the cover-tree tableau. Repaired mode (obligations enforced on actual completed relations; no EQ entries; mandatory propagation): all ten diagnostics correct — $C_{\text{force}}, C_{\text{split}}, C_{2\text{hop}}, C_{3\text{hop}}, \exists \text{DR}. A \sqcap \forall \text{DR}. \neg A$ UNSAT; $C_{\text{recursive}}, C_{\uparrow}$, PO-loop, the round-10 dual-descendant tower concept, and a sanity SAT concept SAT. Literal mode (either loophole active): **ACCEPTs the four unsatisfiable forced-composition concepts**. Random sweep: 148 concepts, zero disagreements. Caveats: bounded occurrence budget; regularity via request-closed blocking (Finding 7).

6 The repair program

The missing keystone is now one lemma. Two routes, both compatible with the manuscript’s architecture; route (a) is recommended.

(a) The discipline route (recommended)

- (R1) **Vertical liveness.** Extend saturation/(V7) to the shadow layer: a pair whose relation is (or becomes, by forced composition) PP, PPI or EQ must be *co-bagged* (live-live), never shadow-carried. Since the only compositions forcing a vertical atom are PP; $PP = \{PP\}$, $PPI; PPI = \{PPI\}$ (and EQ-trivial ones), forced verticals arise only along containment chains, which the presentation already keeps live; the WP15 counterexample (which needs PP/PPI shadow entries) becomes ill-formed. Shadow entries then range over $\{DR, PO\}$.
- (R2) **Horizontal Shadow Amalgamation Lemma** (to be proved): bag networks + DR/PO-only shadow rows satisfying one-point closure and rigidity are jointly realizable. Proof strategy: fix each shadow's extended family (sound alone, by the single-shadow argument / WP15 P3); complete shadow-shadow pairs by DR — every cell $\text{Comp}(r, s^{-1})$ with $r, s \in \{DR, PO\}$ contains DR — and derive triangle-closure through ports from the per-bag one-point closure plus directional propagation along separators. WP15 P4 found no counterexample (358 configurations).
- (R3) **Bookkeeping fixes.** Findings 2, 3, 6(iii): non-EQ entries; mandatory propagation; requests keep their source live along the support branch.

(b) The extended-bag route

Promote in-scope shadows to *virtual ports*: each bag carries a complete atomic closed network on $P \cup \text{Sh}$ including shadow-shadow entries, with boundary agreement on continued shadows. The family is then literally a tree-shaped agreeing family and Theorem 2.5 applies verbatim (WP15 P5: no failures). The cost is real: along an infinite branch the number of simultaneously in-scope shadows is unbounded, so the *alphabet* must quotient shadow instances by profile, and the quotient needs its own lemma — same-profile instances must be pairwise realizable (PP and PO are always available in $\text{Comp}(r, r^{-1})$, but a profile whose own obligations forbid all four non-EQ relations to its own type — e.g. the PO-loop type — is *solitary* and must not be duplicated). This route is doable but heavier than (a).

Also required

Finding 4: witness-generated context accounting (or the 2^{poly} mosaic bound); Finding 5 wording; Finding 6 formalities.

7 Status recommendation

The trajectory is genuinely convergent: each round has replaced a bespoke global-coherence mechanism by a smaller, more standard one, and the residual gap has shrunk correspondingly — from transformer coherence over routes (round 11) to amalgamation of DR/PO shadow rows over a tree decomposition (round 12). The patchwork half of the sixth review's recommendation is now genuinely implemented and machine-corroborated; what remains unproved is the interaction of the *obligation* layer with the amalgam. Accordingly:

- Keep the project status: **strongly supported, NOT certified**; seven reviews, seven defects (this report's Finding 1 continuing the series).

- Name the keystone precisely: the **Shadow Amalgamation Lemma** (with the vertical-liveness discipline), plus the $K(C_0)$ accounting.
- Round-12 remains *not cold-reviewed*; this hot audit does not change that.
- The new probes WP14–WP16 should be listed alongside WP1–WP13; WP14 directly backs the proof’s central external dependency, WP15 delimits the open lemma, WP16 is the first acceptance-level cross-validation of the patchwork-bag mechanism against the cover-tree tableau.

References

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